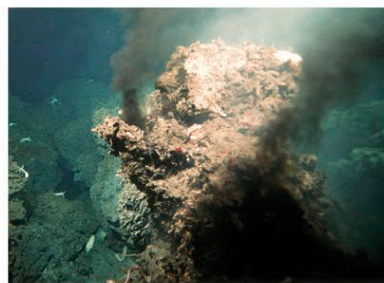


ARCHAEA AND BACTERIA

An alien visiting Earth might conclude that its true masters were the archaea and the bacteria. They outnumber, and are more diverse than, the more complex life forms known as the eukaryotes, and thrive in every nook and cranny of the planet.

DOMAIN	ARCHAEA
PHYLA	12
ORDERS	18
FAMILIES	28
SPECIES	Probably millions

DOMAIN	BACTERIA
PHYLA	About 50
ORDERS	180
FAMILIES	430
SPECIES	Probably millions



Hydrothermal vents at the bottom of oceans are home to various thermophile archaea living at high temperatures.



Staphylococcus bacteria are common agents of food poisoning in humans, forming microscopic clusters in food.

DEBATE

CAULDRON OF LIFE

Many archaea are adapted for extremes; in hot water their fragile DNA is lagged by protective protein. Similiar protein braces the longer chromosomes of eukaryotes (fungi, plants, and animals). So perhaps what began as insulation in primeval pools evolved into a "scaffold" for the extra DNA needed for more complex life.

The archaea and bacteria are single-celled organisms and were the first forms of life on Earth. They form two of life's three fundamental divisions, along with the eukaryotes. While all living cells possess DNA, eukaryote cells contain it within a membrane-bound nucleus, and most also have energy-generating mitochondria. Archaea and bacteria have neither a nucleus nor mitochondria. They are only distantly related to each other, having evolved from separate, as yet unknown genetic origins. Archaea cells are contained by chemically unique membranes overlain by a tough outer cell wall, and within the cell their DNA is often protein-covered. Bacteria cells have very different physical and chemical makeups, particularly in their cell wall. Their different properties make the typical archaea especially suited to harsh habitats, while bacteria are ubiquitous and thrive in every environment.

SMALLEST LIVING THINGS

All archaea and bacteria are tiny, their size measured in microns or micrometres (μm); $1\ \mu\text{m}$ equals one-thousandth of a millimetre. A human hair is about $80\ \mu\text{m}$ thick; most archaea and bacteria are just $1\text{--}10\ \mu\text{m}$ in length. Yet they survive in almost every corner of the biosphere, from the outer atmosphere to deep within Earth's crust, and from ocean depths to inside the human body. Some archaea and bacteria thrive in boiling water or freezing ice, survive radiation, and even live off poisonous gases and corrosive acids. Most get nourishment from dead material, others from infesting living bodies. While some make food in darkness, by using the energy in minerals, others use photosynthesis, using the energy in light to convert carbon dioxide and water into food and oxygen. Although notorious for causing infectious diseases, bacteria are actually essential to human health, with humans depending on their gut bacteria to help break down food and to manufacture essential nutrients. For almost 4 billion years, archaea and bacteria have profoundly influenced Earth's climate, rock formation, and the evolution of other forms of life.

MICROSCOPIC CYANOBACTERIA COLONY >

Although bacteria have single cells, the cells of some, such as cyanobacteria, can join together in spectacular, long filaments.

